In [2]:
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

print("Libraries imported successfully")
```

```
Libraries imported successfully
```

In [3]:
```
df = pd.read_csv("titanic.csv")
print("Dataset Loaded successfully")
```

```
Dataset Loaded successfully
```

In [4]:
```
df.head()
```

Out[4]:

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500

In [5]:
```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype  
---  ------       --------------  -----  
 0   PassengerId  891 non-null    int64  
 1   Survived     891 non-null    int64  
 2   Pclass       891 non-null    int64  
 3   Name         891 non-null    object 
 4   Sex          891 non-null    object 
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64  
 7   Parch        891 non-null    int64  
 8   Ticket       891 non-null    object 
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    object 
 11  Embarked     889 non-null    object 
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

In [6]:
```
print("Number of rows:", df.shape[0])
print("Number of columns:",df.shape[1])
```

```
Number of rows: 891
Number of columns: 12
```

In [7]:
```
df.describe()
```

Out[7]:

	PassengerId	Survived	Pclass	Age	SibSp	Parch	Fare
count	891.000000	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	446.000000	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
std	257.353842	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	1.000000	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	223.500000	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
50%	446.000000	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	668.500000	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	891.000000	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

In [8]:
```
print(df.isnull().sum())
```

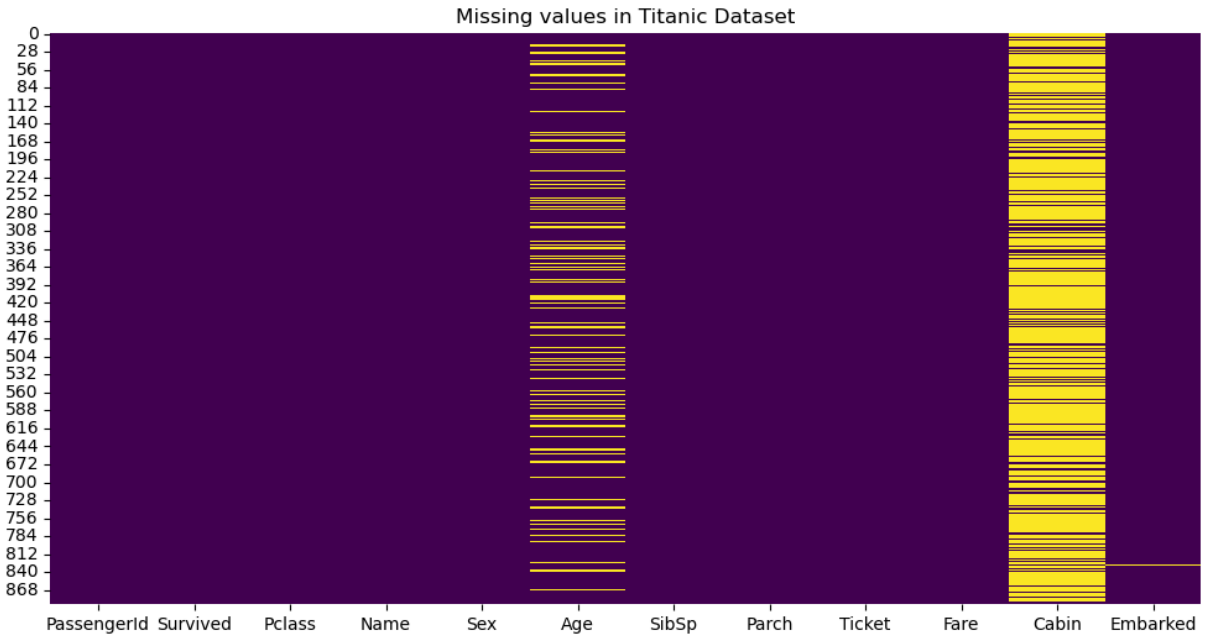
```
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age            177
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin          687
Embarked         2
dtype: int64
```

In [9]:
```
missing_percentage = (df.isnull().sum() / len(df)) * 100
print(missing_percentage)
```

```
PassengerId     0.000000
Survived        0.000000
Pclass          0.000000
Name            0.000000
Sex             0.000000
Age            19.865320
SibSp           0.000000
Parch           0.000000
Ticket          0.000000
Fare            0.000000
Cabin          77.104377
Embarked        0.224467
dtype: float64
```

In [10]:
```
plt.figure(figsize=(12,6))
sns.heatmap(df.isnull(),cbar=False,cmap='viridis')

plt.title("Missing values in Titanic Dataset")
plt.show()
```

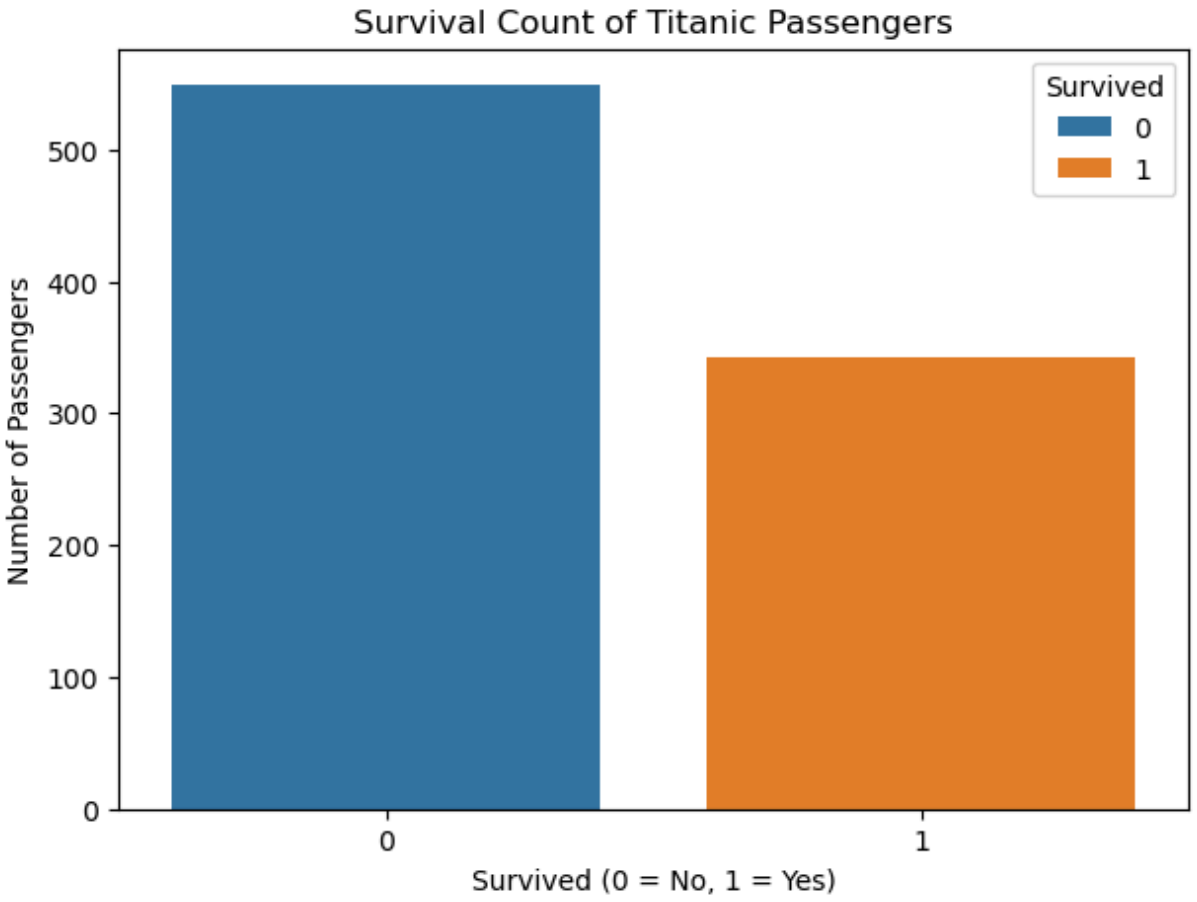


In [11]:
```
plt.figure(figsize=(7, 5))

sns.countplot(data=df, x='Survived', hue='Survived')

plt.title("Survival Count of Titanic Passengers")
plt.xlabel("Survived (0 = No, 1 = Yes)")
plt.ylabel("Number of Passengers")

plt.show()
```

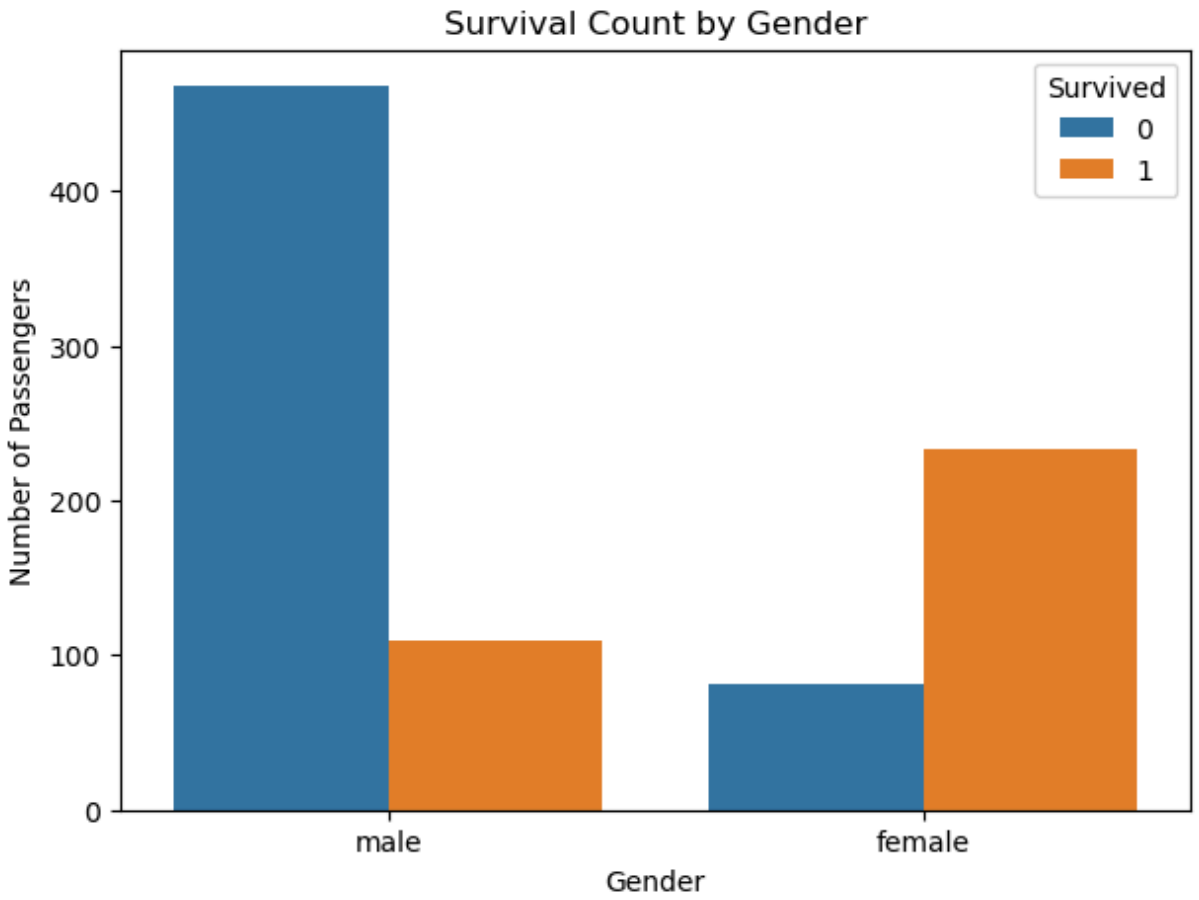


In [12]:
```
plt.figure(figsize=(7, 5))

sns.countplot(data=df, x='Sex', hue='Survived')

plt.title("Survival Count by Gender")
plt.xlabel("Gender")
plt.ylabel("Number of Passengers")

plt.show()
```

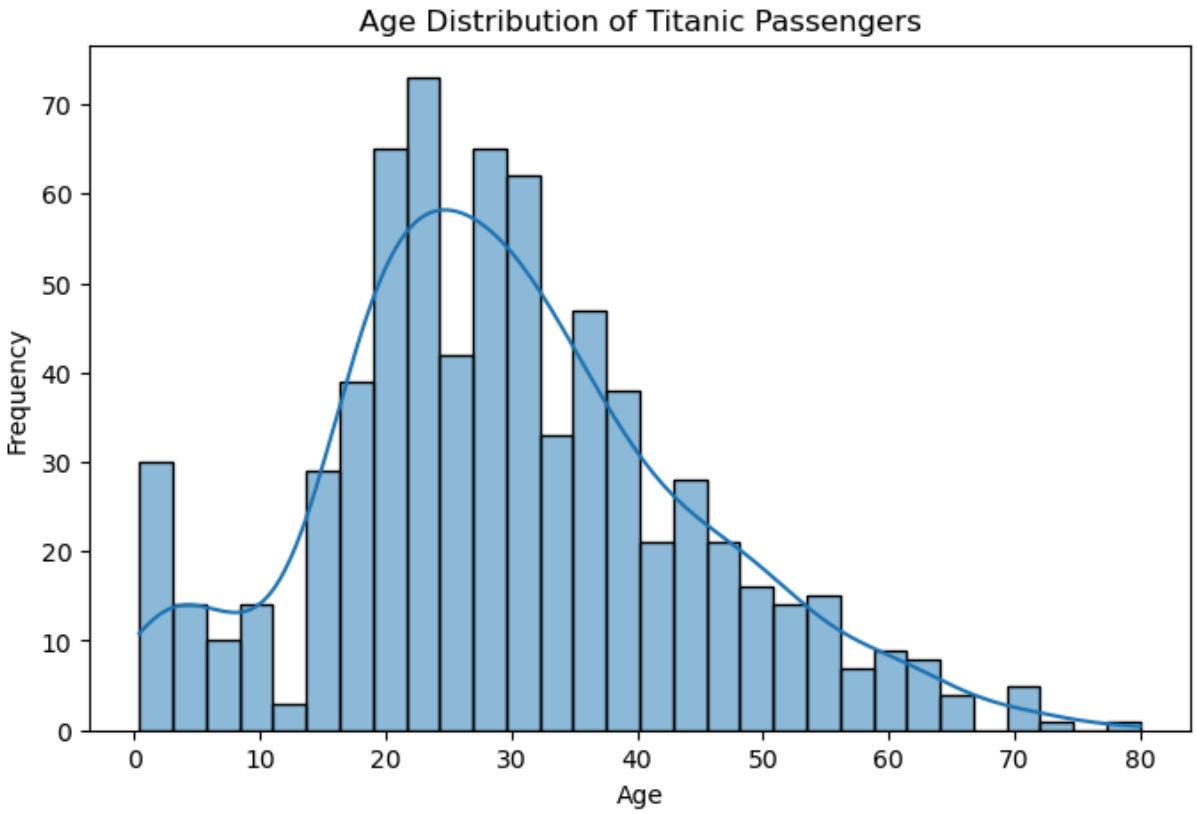


In [13]:
```
plt.figure(figsize=(8, 5))

sns.histplot(data=df, x='Age', bins=30, kde=True)

plt.title("Age Distribution of Titanic Passengers")
plt.xlabel("Age")
plt.ylabel("Frequency")

plt.show()
```

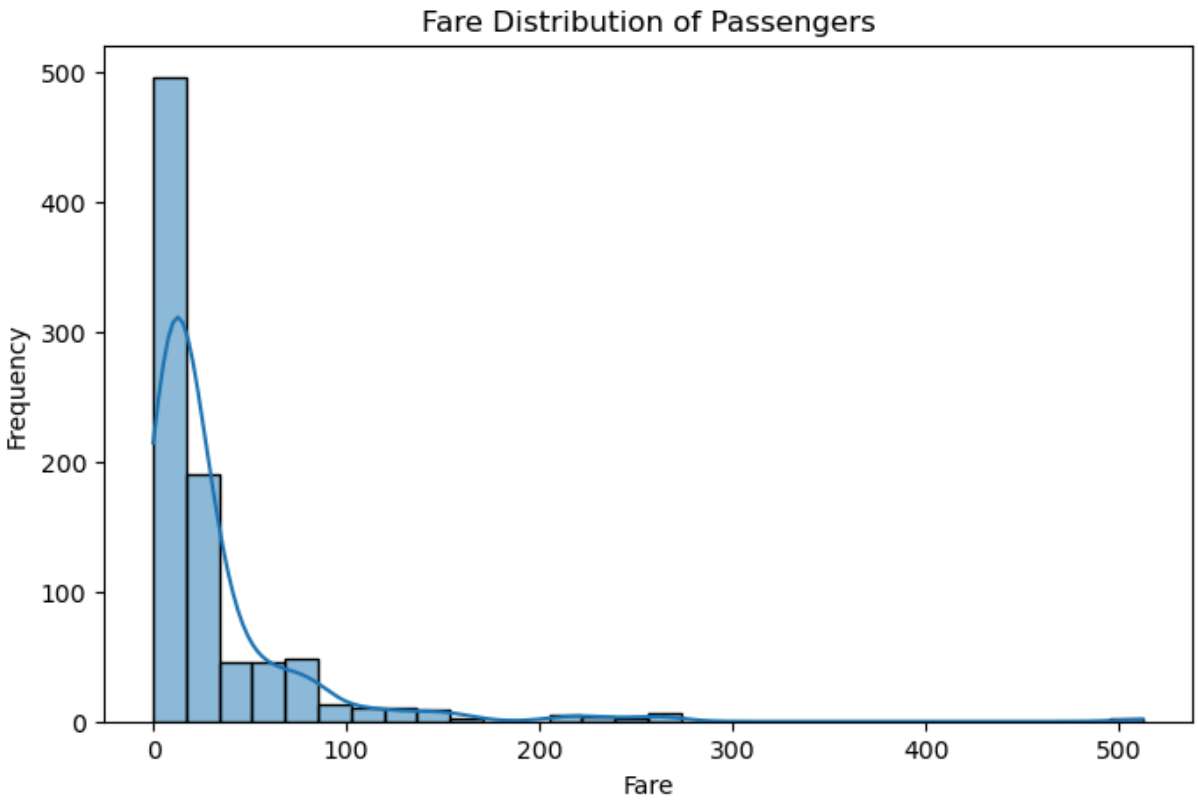


In [14]:
```
plt.figure(figsize=(8, 5))

sns.histplot(data=df, x='Fare', bins=30, kde=True)

plt.title("Fare Distribution of Passengers")
plt.xlabel("Fare")
plt.ylabel("Frequency")

plt.show()
```



In [ ]:
```
df_clean = df.copy()
df_clean['Age'] = df_clean['Age'].fillna(df_clean['Age'].median())

print("Age missing values:", df_clean['Age'].isnull().sum())
```

```
Age missing values: 0
```

In [ ]:
```
df_clean['Cabin'] = df_clean['Cabin'].fillna('Unknown')

print("Cabin missing values:", df_clean['Cabin'].isnull().sum())
```

```
Cabin missing values: 0
```

In [ ]:
```
df_clean['Embarked'] = df_clean['Embarked'].fillna(
    df_clean['Embarked'].mode()[0]
)

print("Embarked missing values:", df_clean['Embarked'].isnull().sum())
```

```
Embarked missing values: 0
```

In [18]:
```
print(df_clean.isnull().sum())
```

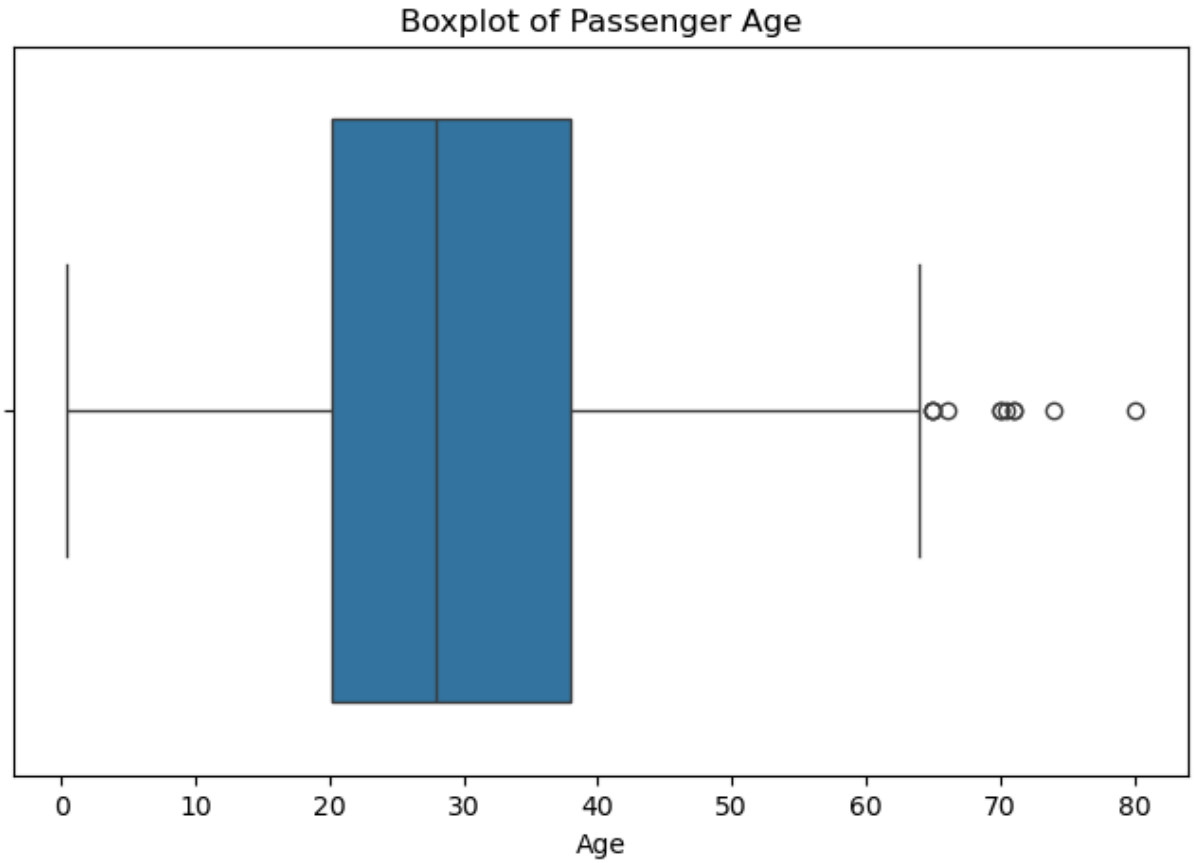
```
PassengerId    0
Survived       0
Pclass         0
Name           0
Sex            0
Age            0
SibSp          0
Parch          0
Ticket         0
Fare           0
Cabin          0
Embarked       0
dtype: int64
```

In [19]:
```
plt.figure(figsize=(8, 5))

sns.boxplot(data=df, x='Age')

plt.title("Boxplot of Passenger Age")
plt.xlabel("Age")

plt.show()
```

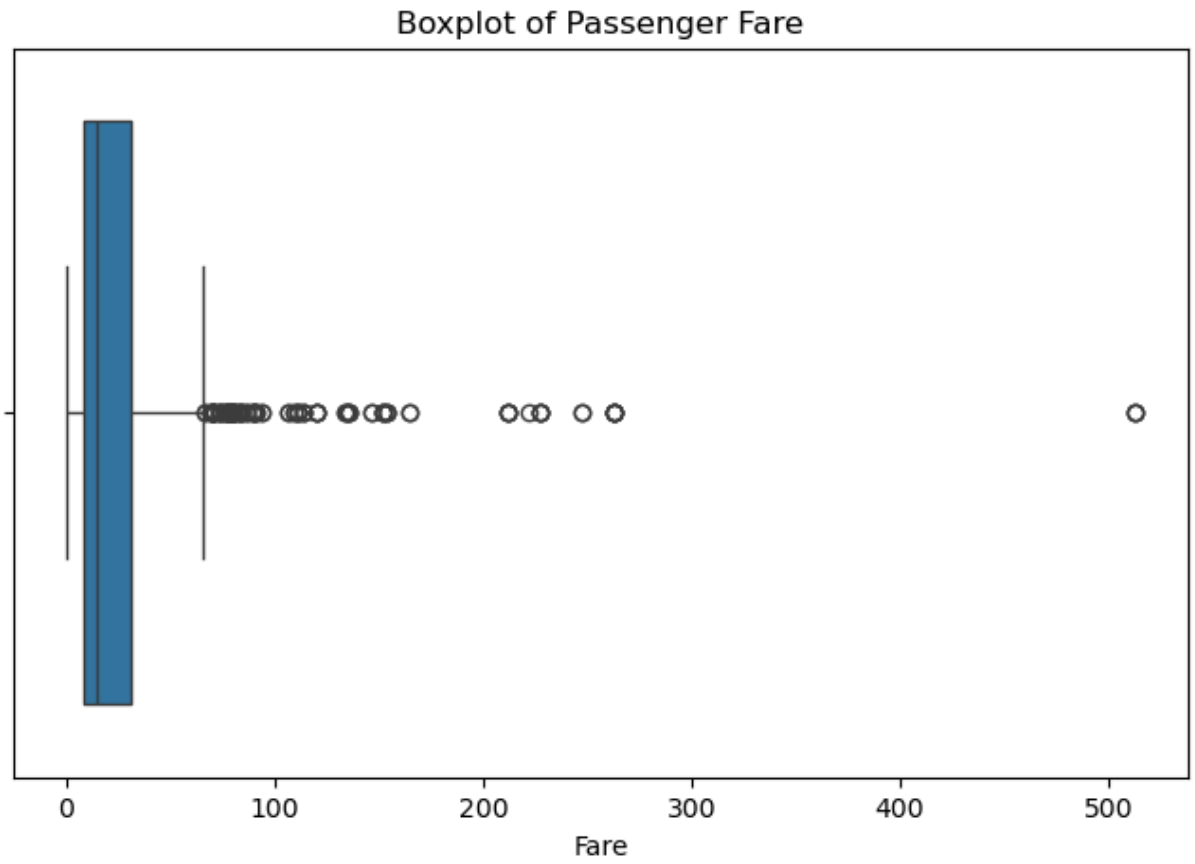


In [20]:
```
plt.figure(figsize=(8, 5))

sns.boxplot(data=df, x='Fare')

plt.title("Boxplot of Passenger Fare")
plt.xlabel("Fare")
```

```
plt.show()
```

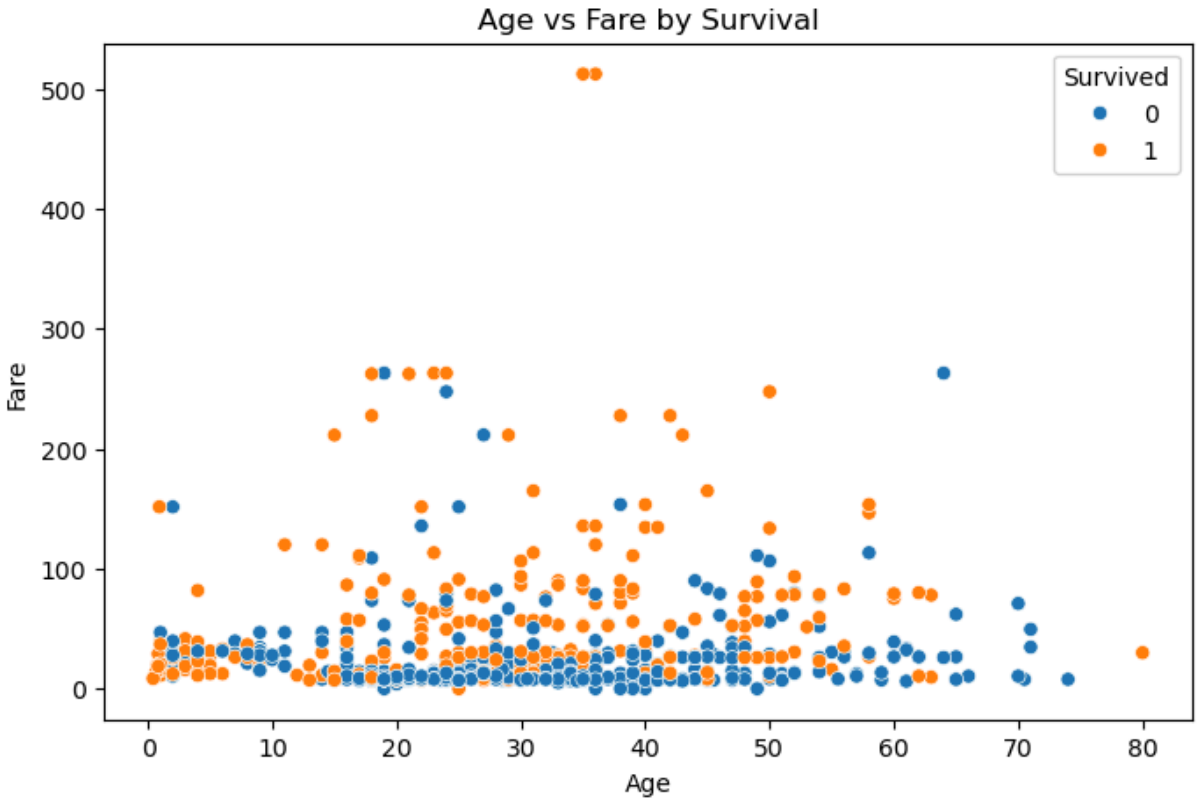


In [21]:
```
plt.figure(figsize=(8, 5))

sns.scatterplot(data=df, x='Age', y='Fare', hue='Survived')

plt.title("Age vs Fare by Survival")
plt.xlabel("Age")
plt.ylabel("Fare")

plt.show()
```

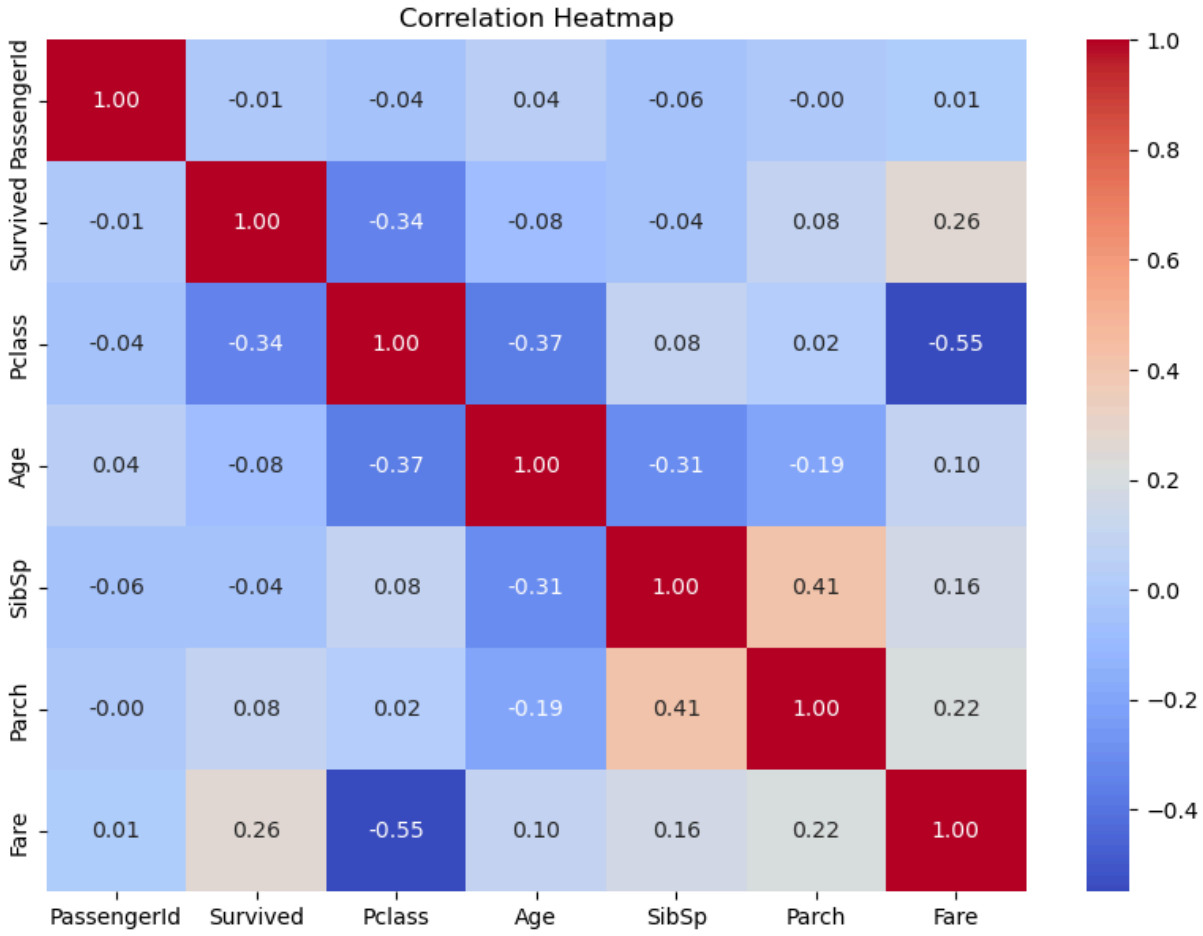


In [22]:
```
plt.figure(figsize=(10, 7))

sns.heatmap(df.select_dtypes(include='number').corr(),
            annot=True,
            cmap='coolwarm',
            fmt='.2f')

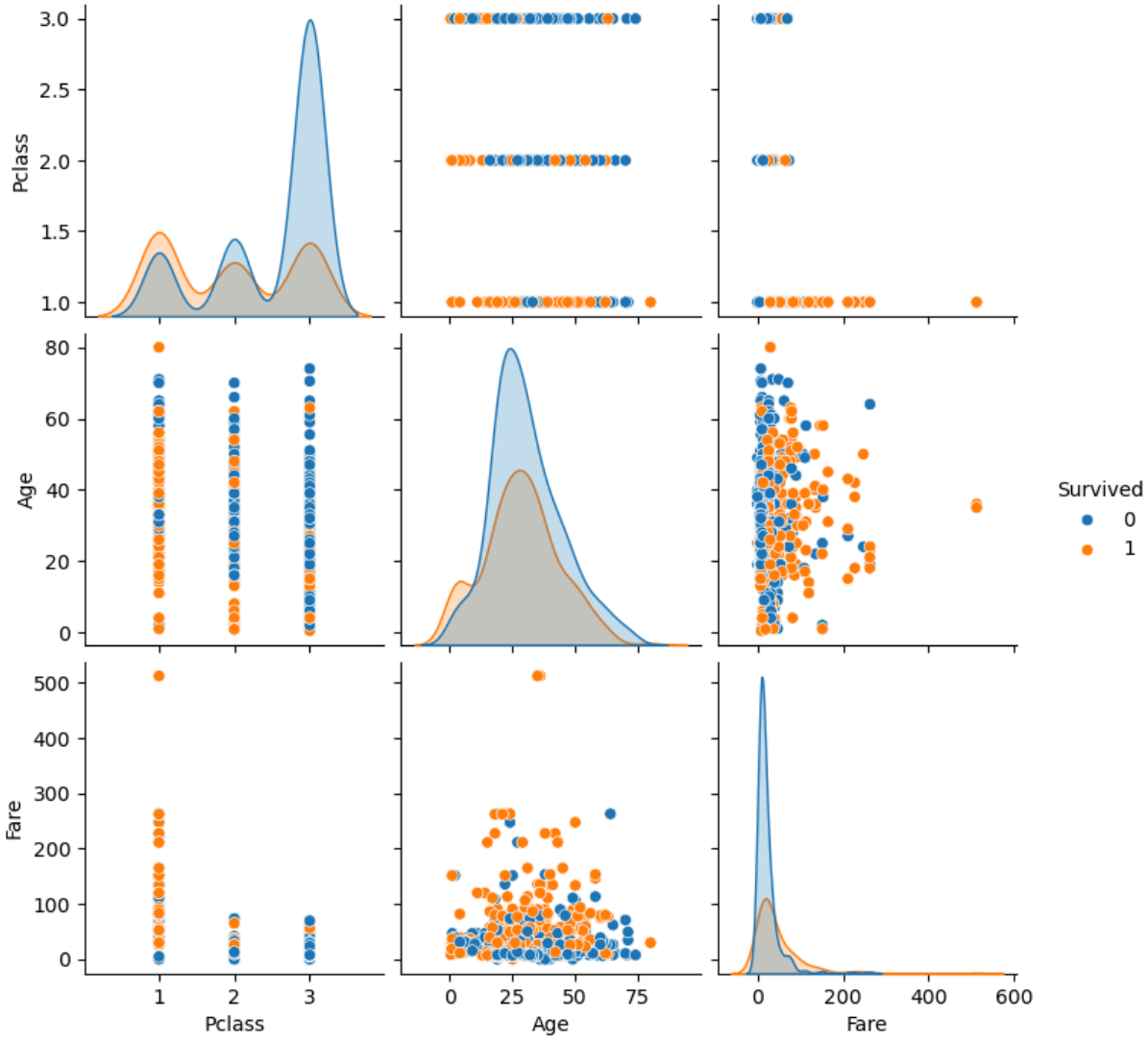
plt.title("Correlation Heatmap")

plt.show()
```



In [23]:
```
sns.pairplot(
    df[['Survived', 'Pclass', 'Age', 'Fare']],
    hue='Survived'
)

plt.show()
```



In [24]:
```
df_clean.to_csv("titanic_cleaned.csv", index=False)

print("Cleaned dataset saved successfully!")
```

```
Cleaned dataset saved successfully!
```

In [25]:
```
# Final cleaned dataset overview

print("Original Dataset Shape:", df.shape)
print("Cleaned Dataset Shape:", df_clean.shape)

df_clean.info()
```

```
Original Dataset Shape: (891, 12)
Cleaned Dataset Shape: (891, 12)
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype  
---  ------       --------------  -----  
 0   PassengerId  891 non-null    int64  
 1   Survived     891 non-null    int64  
 2   Pclass       891 non-null    int64  
 3   Name         891 non-null    object 
 4   Sex          891 non-null    object 
 5   Age          891 non-null    float64
 6   SibSp        891 non-null    int64  
 7   Parch        891 non-null    int64  
 8   Ticket       891 non-null    object 
 9   Fare         891 non-null    float64
 10  Cabin        891 non-null    object 
 11  Embarked     891 non-null    object 
dtypes: float64(2), int64(5), object(5)
memory usage: 83.7+ KB
```

In [26]:
```python
print("Total Passengers:", len(df_clean))

print("\nSurvival Rate:")
print(df_clean['Survived'].value_counts(normalize=True) * 100)

print("\nGender-wise Survival Rate:")
print(df_clean.groupby('Sex')['Survived'].mean() * 100)

print("\nClass-wise Survival Rate:")
print(df_clean.groupby('Pclass')['Survived'].mean() * 100)

print("\nAverage Age:", df_clean['Age'].mean())

print("\nAverage Fare:", df_clean['Fare'].mean())
```

```
Total Passengers: 891

Survival Rate:
Survived
0    61.616162
1    38.383838
Name: proportion, dtype: float64

Gender-wise Survival Rate:
Sex
female    74.203822
male      18.890815
Name: Survived, dtype: float64

Class-wise Survival Rate:
Pclass
1    62.962963
2    47.282609
3    24.236253
Name: Survived, dtype: float64

Average Age: 29.36158249158249

Average Fare: 32.204207968574636
```

Exploratory Data Analysis on Titanic Dataset

1. Introduction

Exploratory Data Analysis (EDA) is used to understand data patterns, identify missing values, detect outliers and discover relationships between variables.

2. Dataset Description

The Titanic dataset contains passenger information such as Age, Gender, Passenger Class, Fare and Survival Status.

3. Data Cleaning

- Checked missing values.
- Filled missing Age values using Median.
- Filled missing Cabin values with Unknown.
- Filled missing Embarked values using Mode.
- Created a cleaned dataset.

4. Data Visualization

The following visualizations were performed:

- Countplot
- Histogram
- Boxplot
- Scatterplot
- Correlation Heatmap
- Pairplot

5. Outlier Detection

Used Boxplots and the IQR method to identify outliers in the Fare column.

6. Key Findings

Analyzed passenger survival rates, gender-wise survival, passenger class, average age and average fare.

7. Conclusion

EDA helped understand passenger characteristics, survival patterns, missing values and relationships between different variables.